

Exam Preparation: Hardware and IO Allocation (DCS)

As of 17 Sep 2026 · Martin Feuerstein

Overview

The exam consists of **20 randomly selected questions in 25 minutes** (Moodle, in the classroom, week 39; open 22-24 Sep 2026) and is worth 20 points. Moodle and EAS materials are allowed — a well-structured cheat sheet therefore saves time.

Exam topics according to Moodle:

1. Valmet DNA hardware
2. Station names and roles
3. System diagram (system layout)
4. IO address allocation
5. FbCAD and DNA Explorer — basic functions
6. Download procedures and Function Test tool

Example questions from the announcement: key feature of DCS vs. PLC · what DCS stands for · what GB01 indicates in the naming policy · role of the backup station · what the black wire colour represents in the DNA software · what happens if two blocks have the same tag name · what the Function Test tool allows you to do.

Sources: Moodle sections 1 (Hardware engineering), 2 (I/O application engineering) and their attachments (course files 1_ to 5_).

1. DCS in general (DCS vs. PLC)

DCS = Distributed Control System. PLC and DCS perform the same core task (read inputs → execute the application → write outputs); the difference lies in organisation, scale and lifecycle.

Aspect	Typical PLC	Typical DCS
Application	Machines, compact automation (packaging, robot cell)	Large continuous processes (power plant, pulp mill, refinery, water)
Size	Usually a smaller I/O count	Large I/O count (thousands of signals)
Operation	Often runs autonomously, local HMI	Continuous operator interaction (setpoints, modes, alarms, trends)
Architecture	One or a few controllers	Functions distributed across specialised stations/nodes
Expansion	Depends on the platform	Scales by adding stations, nodes and subsystems
Engineering	Separate projects per area	Central, integrated engineering (EAS)

Five key ideas from the lecture:

1. PLC and DCS perform the same fundamental control task.
2. A DCS is designed for integrated, plant-wide process automation.
3. Continuous operator interaction is a central DCS requirement.
4. Distributed, specialised components improve scalability and management.
5. Long lifecycle (decades) and gradual modernisation shape the architecture.

Concept comparison PLC → DCS: PLC CPU → process station · TIA Portal project → central engineering environment · HMI/SCADA → operator station · device diagnostics → integrated system diagnostics · distributed I/O and fieldbus → distributed process I/O and fieldbus.

Valmet DNA history: Damatic (1979) → Damatic XD (1988) → XDi → metsoDNA (2000) → Metso DNA → Valmet DNA (2015). Application compatibility since 1988. Other vendors: Siemens PCS 7, ABB 800xA, Honeywell Experion, Emerson DeltaV.

Scalability: compact (controller + I/O + server functions in one industrial computer) · laboratory/medium (dedicated process station + physical I/O, virtualised servers — as at JAMK) · large (several subsystems, process stations, control rooms).

2. Stations / nodes and their roles

In Valmet DNA the functions are distributed across specialised stations connected by **redundant Ethernet** (Valmet-specific protocol over **UDP/IP**, name-based communication).

Station	Code	Operating system	Task
Process Control Station	PCS (P)	Modified Linux	Executes the control applications in real time (FbCAD logic, SeqCAD sequences, PID, MPC, fuzzy) and connects field devices through the FBC
Backup Station	BU (B)	Windows	Not a backup in the literal sense: distributes the system software during download, holds all node applications, nodes retrieve their applications/configuration from the BU after booting, enables automatic recovery after a power failure, central version management
Alarm Station	ALS/ALP (A)	Windows	Collects alarms from all components, timestamps, categorises and prioritises them (hierarchy) and distributes them to the operator stations; state synchronisation
Engineering Activity Server	EAS (E)	Windows	Central engineering database (all FbCAD applications, configurations, documentation), multi-user access, lifecycle and version management, tools (FbCAD, SeqCAD, Graphics Designer), function testing , Network Designer (system and cabinet layout)
Operator Station	OPS (O)	Windows	Monitoring and operation with Valmet DNA Operate (displays, alarms, trends)
Diagnostic Station	DIA (D)	Windows	Diagnostics (DIA, Diag@Web, Maintenance@Web)
OPC Station	O	Windows	OPC connections to other systems (e.g. GO01)

FBC = Field Bus Controller — the interface from the PCS to the field devices: ACN IO / MIO (rack I/O), PROFINET, Profibus, OPC/OPC UA, Modbus RTU/TCP.

ACN = Application and Control Node — hardware versions of the PCS:

- **ACN MR** (Modular Rail; deprecated ACN SR1): small form factor, limited scalability → **GP01 in lab DP66**
- **ACN CS** (deprecated ACN C20): industrial PC, better scalability → EP01/EP02 in lab DP39
- **ACN RT**: industrial PC, most powerful option

BU/ALS/OPS run on ACN AS, ACN PO, ACN CSW — or **virtualised**, as at JAMK. Nodes can be combined freely (e.g. GB01, G109 and G1A1 on one VM) → flexibility.

3. Naming policy and slots

A station name consists of **subsystem + (optional control room) + station type + number**.

Character	Meaning	Values
1st character	Subsystem (plant area)	G = automation lab DP66, E = energy lab DP39 (generally A, B, C ...)
optional digit	Control room within the subsystem	G1, G2, E1 ...
Type letter	Station type	P = process controller station (PCS) · B = backup (BU) · A = alarm (ALS) · E = EAS · O = operator (OPS) or OPC · D = diagnostics (DIA)

Character	Meaning	Values
Number	Instance within the subsystem	01, 09, 1 ...

Examples (exam relevant):

- **GP01** = subsystem G, process controller station, no. 01 (the PCS in lab DP66, ACN MR)
- **GB01** = subsystem G, **backup station**, no. 01
- **G1A1** = subsystem G, control room G1, alarm station no. 1
- **G109** = subsystem G, control room G1, operator station no. 9
- **EB01** = subsystem E, backup station no. 01
- **EAS-G** = Engineering Activity Server of subsystem G
- **GO01** = OPC server, **EP01/EP02** = process stations in the energy lab

Node address: every station has a hexadecimal node/token address (e.g. GP01 = 6161, EP01 = 4161, EP02 = 4181; VM GB01/G109/G1A1 = 64a1).

Slots: in the days of Damatic XD and the VME bus a station consisted of physical cards, each with its own function (CPU, NCU, FBC, BU ...). Today slots still exist, but **virtualised** ("Slot Controller — DNA Program Starter"): each slot starts a program.

- On the Windows VM: slot 01 = BU program (GB01), slots 03/04 = Operate (G109), slot 06 = alarm program (G1A1), slots 15/16 = NCU2 (network communication).
- On GP01: slot 1 = PCS program (GP01), further slots = FBC programs (fbcl) and IBC, **slot 2 = ACN IO (FBC 2)**, **slots 4, 5, 6 = Profibus DP** (GP01_m4 ... m6), **slot 9 = PROFINET**, NCU2 for communication.

The node structure is managed in the **System Configurator** (subsystems → nodes → functions/slots).

4. Hardware and system diagram

In lab DP66 the process station **GP01 (ACN MR)** with local **ACN IO** (M80 series) controls the water process; all servers run virtualised on a Dell server (VDNA1ESXI1) in the server room.

JAMK system layout

Area	Components
G system, automation lab DP66	Thin client G-THIN1, I/O cabinet with GP01 (ACN MR, 192.168.45.62, node 6161), Profibus coupler GP01-PB (m4-m6), PROFINET switch GP01-9-SW10
E system, energy lab DP38/39	Thin clients E-THIN1/2, PC cabinet with EP01/EP02 (ACN CS, netboot, nodes 4161/4181)
Server room BK10 (IT cabinet)	VMware host VDNA1ESXI1 with the VMs: EAS-G (GD00, GDS1, GWS1), EAS-E, G109 VM (GB01 + G109 + G1A1: BU, RDS, ALS, Loop AutoTuner), GO01 (OPC server), SIMU (paper machine simulation), E-INFO (historian)
Network	JAMK network AUTOM_EL, VLAN121, 192.168.45.0/24, gateway .254; redundant Ethernet, star or ring topology

Capacity of the automation lab: EAS 31 simultaneous users, RDS 35 operators, OPC connections to other systems.

PCS hardware (ACN MR / GP01)

- **FBC 2** = ACN IO (local rack I/O) · **FBC slots 4, 5, 6** = Profibus DP · **slot 9** = PROFINET
- **IBC** = I/O bus controller (I/O group, number 0-15) · **IPSP** = power supply · **MB8** = mounting base for 8 units
- One I/O unit holds up to **16 I/O cards** (card place 0-15)

Card layout in cabinet DNA01 (GP01, IBC 0):

Card place	0	1	2	3	4	5	6	7
Card	DI8P	DI8P	DO8P	DO8P	AI8C	AO4C	AO4C	free

ACN I/O units (M80 series)

Card	Channels	Signal	Key features
DI8 (DI8P / DI8N)	8	Digital input 24 V (PNP / NPN)	Channel-specific current-limited supply (40 mA); state 1 = I > 4.5 mA, state 0 = I < 2 mA; measuring interval 1 ms; also pulse counter mode
DO8P	8	Digital output PNP 17–30 V DC	Mechanical relay (normally open), < 200 mA; parameterisable safe state (no change / 0 / 1) after 1 s–4 h
AI8C / AI8V	8	0/4...20 mA or 0/2...10 V	AI8C has its own 24 V transmitter supply per channel; 16 bit; monitors short circuit, wire break, range over/undershoot; measuring interval 4.4 ms
AO4C / AO4V	4	0/4...20 mA or 0/2...10 V	14 bit; monitors wire break and short circuit; update interval 5 ms
TI4 (TI4W3 / TI4W4)	4	Pt100, 3- or 4-wire	Rated range –50...+400 °C; 3-wire compensates the wire resistance (assuming identical wires), 4-wire is independent of it

Wiring (traditional, wired I/O)

Field device → **device cable** (usually 2-pair) → **field box** (close to the device) → **trunk cable** (usually 24-pair instrumentation cable) → **cross connection** in the I/O cabinet (terminals XW/XS) → flat cable → **cross connection board CXW/CXS/CXR** → I/O card. The CX pinout is in the manual of the I/O card, the connection data of the field device in its own manual.

HW engineering workflow

1. Collect the initial data (basic engineering: PI diagram, system description)
2. Select the field devices (signal, range, supply, current)
3. Prepare the **loop list and the IO list**
4. Instrumentation loop diagram: connections at the device and in the field box, cabling, cross connection, connection to the I/O cards

5. IO address allocation and IO list

An ACN I/O address has four parts: **FBC slot : IBC number : card place : channel** — e.g. **2:0:4:0** = FBC 2, IBC 0, card 4 (AI8C), channel 0.

Parameter in the I/O module	Range	Meaning
\$FBC	2–15	Slot of the field bus controller (ACN IO on GP01 = 2)
\$PIC (IBC number)	0–15	I/O group / I/O bus controller (GP01 = 0)
\$CARD (card place)	0–15	The unit's place in the I/O group
\$CHANNEL	0–7	Channel on the card (DI8/DO8/AI8: 0–7, AO4/TI4: 0–3)

Example from the "IO application" assignment (loop PARAAN-ACN-IO):

Module	Card	Address	Function
pr:xxxx-HS-100.I	DI8	2:0:0:0	Switch in the field box
pr:xxxx-HS-101.I (second DI, name assumed)	DI8	2:0:0:1	Same card, channel 1
pr:xxxx-GX-100.O	DO8	2:0:2:0	Signal light, red
pr:xxxx-GX-101.O	DO8	2:0:2:1	Signal light, yellow

Module	Card	Address	Function
pr:xxxx-TT-100.I	AI8	2:0:4:0	Temperature measurement Pt100 (via 4...20 mA transmitter)

Further AI parameters: minimum/maximum (scaling), range (0-15, e.g. 4...20 mA), filter, line fault high/low limits.

IO list

The IO list assigns every field signal to exactly one channel. Columns of the course template: **process station · FBC # · IBC # · card type · card place · channel · device tag · position · signal · operation · manufacturer.**

- Example rows: GP01 · 2 · IBC 0 · DI8 · 0 · 0 · LS-1 · LZ-1 · 24 VDC · overflow switch; GP01 · 2 · IBC 0 · AI8 · ... · LT-1 · LI-1 · 4...20 mA · tank level
- Take the **device tag** (device in the field, e.g. LT-1, LS-1, SC-1) and the **position** (function in the system, e.g. LI-1, LZ-1, SIC-1) from the PI diagram
- Assumption of the assignment: max. 16 units per I/O unit
- Procedure: count the signals per device (DI/DO/AI/AO) → total → required cards (DI8/DO8/AI8 = 8 channels, AO4 = 4 channels) → assign the channels; use different colours per card

Water process (PI diagram DP66): tanks T1-T3, pumps P-1/P-2 (speed SC-1/SC-2 via frequency converter, 4...20 mA), level LT-1...LT-3, level switches LS-1...LS-3 (protection LZ), control valves LV-1/LV-2 (controller LIC-1), flow FT-1/FT-2, shut-off valves HV-1...HV-3 (24 V DC).

6. FbCAD, DNA Explorer, download and Function Test

FbCAD (Function Block CAD) is the graphical tool for the control logic; **DNA Explorer** is the program for browsing the Valmet DNA software (engineering database on the EAS). Access by remote desktop: remote.jamk.fi → Apps → **EAS-G** (water process/DP66) or EAS-E (energy lab). **Always sign out** afterwards to release the licence.

Administration part of an FbCAD diagram

Attribute	Meaning
Name	pr:POSITION.F — name of the function module, unique within the system, always starts with pr:
Tag	POSITION (without pr: and .F) — the unique ID under which the loop is stored in the engineering database
Loop name	Comment, shown in DNA Explorer
Package	Name of the process station the loop runs on (GP01)
Execution / Order	Cycle time (e.g. 1000 ms) and execution order (e.g. 20)
Department	Process areas → determines the place in DNA Explorer's process area hierarchy

Same tag = overwrite: if an already existing tag is used, **the new loop overwrites the old one.**

Structure of a diagram

- Left field: external inputs and **I/O input modules** (.I) · middle: function blocks · right field: external outputs and **I/O output modules** (.O) · above: control room modules (position, operation, event)
- **Direct Access Port** (Common → Direct Access Port): global variable, known in the entire DNA system → **the name must be unique system-wide**; used for the connection to the operating display. am, mgv, mtr and pid already include a direct access port.
- **Interfaces in/out** (Common → Interfaces): connection point other loops can read from; the name is used only once per diagram (:out1), it is not global.
- References: direct access port member pr:MITTAUS:hha · port with a single variable pr:MITTAUS.B1 · interface out pr:MITTAUS.F:out1

- Important blocks: **am** (analog measurement, alarm limits HH/H/L/LL with hysteresis, often 1 %), **mtr** (motor), **mgv** (magnetic valve), **pid** (controller), **ccoX** (conditional copy, mode 0 = copy while cnd = 1, 1/2 = rising/falling edge)
- Always change values via **Edit → Values** or the pencil icon, **never** by double-clicking in the AutoCAD dialog.

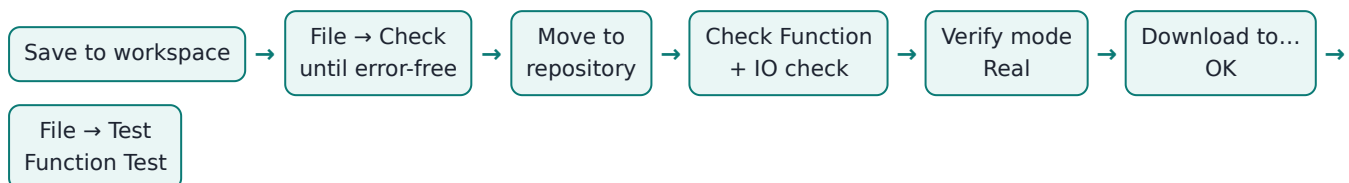
Wire colours (signal types)

With **Draw → Scalar autocolor wire** the line automatically takes the colour of its start point.

Colour	Type
Black	bin — binary signal
Green	ana — analog signal (fault bits + float)
Dark green	anaev — analog with time stamp
Yellow	float — floating point number
Red	fails — fault bits
Violet	bo — binary output (bv + pulse length pw)
Blue / light blue	intl / ints — long / short integer
Brown	binev — binary with time stamp
Grey	general type

Structured signals (e.g. ana, bo) are split with the **Specifier** command, e.g. `:bv` from the bo output of the mtr block.

Checking and download (procedure in the lab)



Step by step:

1. Save the diagram to the **workspace** (File → Save As) and run **File → Check** until the check log reports no errors.
2. Move loops with real I/O modules to the **repository** before loading (DNA Explorer → Move to Data Storage → Repository) and switch the working directory to Repository.
3. **Check Function** with the check scope internal, identifier, external and **IO address**.
4. Make sure **Real** (not Virtual) is selected.
5. **Download to...** → Modular Download: command **Add** (new) or **Replace/Update**, option **Accept to backup** → OK.
6. Only download when you are **physically present in the lab**.

Real vs. Virtual: Download Real works only from the repository and checks internal + identifier + **IO addresses**; Download Virtual works from repository or workspace and checks without IO addresses. Only the configuration functions are transferred (continuous control → PCS, operation → operation server, events → alarms & events server). Up to 5 versions can be kept in the repository.

Function Test tool

Function Test tests a function block or sequence diagram with **continuously updated values from the runtime environment (real or virtual)**.

- Start: FbCAD → **File → Test**, or DNA Explorer → Design → Test Function; the diagram must be downloaded to the PCS first
- Insert **test points** (one by one, by area, whole page) → blue values at the connection points; active fault bits appear as a sum, e.g. `3.1415+(136)` = Dis (128) + Sex (8)
- **Enter new values** for test points and block parameters, several values at once via tables of values

- **History:** record values, pause, step back and forth in time, trend curves
- Test points are **not saved**; Function Test **does not change the original diagram**
- Return with "Return to CAD"

Valmet DNA manuals for reference

The course material is enough for the exam; these manuals provide the details if a question goes deeper. In the lab all manuals are under Start → Valmet DNA → **DNA Manuals** (Application Libraries → PCS Function Blocks).

Manual	File	Use in this exam
Function Block CAD Manual	g5038.pdf	Menus (File → Check/Test/Download Real/Virtual), ACN I/O M80 modules (ch. 7.9.1), wire colour codes (ch. 7.4.1)
Function Test Manual	g5104.pdf (in the course as "Function Test tool manual")	Test points, entering values, history, trends
Hardware CAD Manual	g5105.pdf	Hardware documents on the EAS: structural layout, cabinet and I/O cabinet layouts, circuit diagrams, cable lists, wiring tables, labels
Field CAD / Loop Circuit CAD	g5101.pdf / g5108.pdf	Field wiring and loop diagrams
Automation Language (in the course)	-	Signal types, structure of an automation module, external data points
ACN I/O Units, M80 Series (in the course as separate sheets)	-	Data sheets AI8, AO4, DI8N, DO8P, TI4 with CXW/CXS/CXR pinout

Glossary

Term	Meaning
DCS	Distributed Control System
ACN	Application and Control Node (station hardware)
ACN MR / CS / RT	Modular Rail (GP01) / industrial PC (EP01, EP02) / most powerful option
PCS	Process Control Station — executes the control applications (Linux)
BU	Backup station/server — distributes software and applications, recovery
ALS / ALP	Alarm station — central alarm management
EAS	Engineering Activity Server — engineering database and tools
OPS	Operator Station (Valmet DNA Operate)
RDS	Remote desktop server for operators
FBC	Field Bus Controller — fieldbus/I/O interface of the PCS (slot)
IBC	I/O bus controller — I/O group within one FBC
NCU	Network communication (slot program NCU2)
MIO / ACN IO	Valmet rack I/O (M80 series)
CXW / CXS / CXR	Cross connection boards between flat cable and terminals
Workspace / Repository	Working area for drafts / released state for download with real I/O
Tag / Position	Unique identifier of a loop (e.g. LI-1)
Device tag	Identifier of the field device (e.g. LT-1)
Direct Access Port	Global variable, unique in the whole system
Package	Process station the diagram runs on

Term	Meaning
Loop list / IO list	List of all control loops / assignment signal → channel

Exam checklist: example questions with short answers

Question	Short answer
What does DCS stand for?	Distributed Control System
Key feature of DCS compared to PLC?	Integrated, plant-wide automation with functions distributed across specialised stations and continuous operator interaction; scalable, long lifecycle
What does GB01 indicate?	Subsystem G, backup station , number 01
Role of the backup station?	Distributes the system software during download, holds all node applications (nodes load them after booting), automatic recovery after a power failure, central version management — not a backup in the literal sense
What does a black wire mean?	Binary signal (type bin)
Two blocks / loops with the same tag?	The new loop overwrites the old one in the database; direct access port names must be unique system-wide
What does the Function Test tool allow?	Testing a diagram with live values from the (real/virtual) runtime environment: show test points, change values and parameters, view history and trends
What do GP01 / G1A1 / G1O9 mean?	Process station 01 / alarm station 1 in control room G1 / operator station 9 in control room G1
Operating system of the PCS?	Modified Linux (BU, ALS, EAS, OPS: Windows)
Which slot is ACN IO on GP01?	FBC 2 (Profibus 4–6, PROFINET 9)
Address 2:0:4:0?	FBC 2, IBC 0, card place 4, channel 0
Channels of DI8/DO8/AI8/AO4/TI4?	8 / 8 / 8 / 4 / 4
How many cards fit in one I/O unit?	16
What is checked before the download?	File → Check or Check Function (internal, identifier, external, IO address)
Workspace or repository before a Real download?	Repository , mode Real
What is a direct access port?	Global variable, known in the whole DNA system, unique name, connection to the control room
How are wires coloured automatically?	Draw → Scalar autocolor wire (colour taken from the start point)
Name vs. tag attribute?	Name = pr:POSITION.F ; tag = POSITION (without prefix/suffix)
What is "package"?	The process station the loop runs on (GP01)
Where are the applications stored?	In the engineering database on the EAS (DNA Explorer)

Pitfalls:

- The letter O is used for both operator station and OPC station (G1O9 = operator, GO01 = OPC).
- The backup station is not a data backup server.
- A virtual download does **not** check IO addresses, a real download does.
- Channels and card places are counted **from 0**.
- Test points in Function Test are not saved, and the test does not change the diagram.

Exam tip: 25 minutes for 20 questions is roughly 1 minute per question. Materials are allowed, but searching costs time — go through these tables once beforehand.

Sources (course folder DCS_Overall-Content-PDF): 1_Welcome, 2_Learning outcomes, 3_ Hardware engineering (dcsInGeneral, Stations/Nodes and naming principles, 2 HW engineering, Connection example, IO-Cabinet example, Valmet DNA short overview, attachments AI8/AO4/DI8N/DO8P/TI4, JAMK systemlayout, G-system-kaappi, IO list, PI diagram), 4_ IO application engineering (DNA Instructions in a nutshell, Signal types, Function Test tool manual, Automation language, Remote Access), 5_ Hardware and IO Allocation Exam; Valmet DNA manuals g5038 and g5105.